

An infinite walk in $\mathbb{N}^{16}$, using only unit steps, with no three collinear points

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Abstract

Is it possible to walk to infinity, avoiding three collinear points, using as steps only the standard unit basis vectors $(0, 0, \dots, 0, 1, 0, \dots, 0)$? We prove this is possible in 16 dimensions.

1 Introduction

Let $k \geq 2$, and consider an infinite walk in $\mathbb{N}^k$ using only steps that are basis vectors of the form $(0, 0, \dots, 0, 1, 0, \dots, 0)$. For which k can one avoid some finite number of collinear points?

For dimension $k = 2$ every such infinite planar walk must have every possible finite number of collinear points. This is the content of the famous Gerver-Ramsey theorem [2, 10, 6]. Also see [7].

For dimension $k = 3$ Gerver and Ramsey proved that there is an infinite walk that avoids $5^{11} + 1$ collinear points. This was recently improved to 189 by Lidbetter [9].

Of course it is impossible to avoid 2 collinear points in such a walk. This raises the natural question: for which dimensions k can one avoid three collinear points? One cannot do it for $k = 3$, as shown by Brown [2], since every walk of 8 unit steps contains three collinear points.

In this paper we solve the problem for $k = 16$. The construction can be given explicitly as a morphism on 16 letters.

The description of such a walk can be given picturesquely as follows: if an infinitesimal

obstruction is placed at each lattice point visited, then it is possible to see all the other lattice points of the walk from every point of the walk.

1.1 A reformulation

The problem of avoiding 3 collinear points can be rephrased in an equivalent way as follows, which interprets the original problem in the terminology of combinatorics on words.

Consider an infinite sequence $(a_n)_{n \geq 0}$ over a finite alphabet $\Sigma_k = \{0, 1, \dots, k-1\}$, and call it *good* if it contains no two consecutive nonempty blocks x, y (possibly of different sizes) with the property that the $|x|_a/|x| = |y|_a/|y|$ for all letters $a \in \Sigma_k$. Here $|x|$ is the length of the block and $|x|_a$ is the number of occurrences of a letter a .

The vector $\psi(x) := (|x|_0, |x|_1, \dots, |x|_{k-1})$ is often called the *Parikh vector* of the word $x \in \{0, 1, \dots, k-1\}^*$. The vector of letter frequencies $\psi(x)/|x|$ is called the normalized Parikh vector. Thus a good sequence has no two consecutive nonempty blocks with the same normalized Parikh vector.

Equality of letter frequencies is sometimes called *weak abelian equivalence*; see the discussion in [1] and [5, Section 8.2]. Accordingly, the forbidden factor xy is sometimes termed a *weak abelian square*. (An ordinary *abelian square* additionally has $|x| = |y|$ and consists of two consecutive words, where the second is a permutation of the first. Avoiding ordinary abelian squares alone is therefore not the whole requirement in this problem.)

Proposition 1. *Let $\Sigma = \{0, \dots, k-1\}$ and let $E_0, \dots, E_{k-1}$ be the standard basis vectors of $\mathbb{N}^k$. Define*

$$P_0 = 0, \quad P_n = \sum_{j=0}^{n-1} E_{b_j} \quad (n \geq 1).$$

The word $(b_n)_{n \geq 0}$ is good if and only if no three distinct vertices of $(P_n)_{n \geq 0}$ are collinear.

Proof. The sum of the coordinates of P_n is n , so all vertices are distinct. Fix indices $\alpha < \beta < \gamma$ and set $p = \beta - \alpha > 0$, $q = \gamma - \beta > 0$. With x, y as in the definition,

$$P_\beta - P_\alpha = \psi(x), \quad P_\gamma - P_\beta = \psi(y).$$

If the three vertices are collinear, their two displayed nonzero displacements are scalar multiples: $\psi(y) = \lambda\psi(x)$. Taking coordinate sums gives $q = \lambda p$, so $\lambda = q/p > 0$. Consequently $\psi(x)/p = \psi(y)/q$. Conversely, this equality makes the two displacements positive scalar multiples, so the three vertices are collinear. This holds for every ordered triple of indices, proving both directions. $\square$

For avoidance of triples, Cambie and Kalviainen [4] recently gave a different construction in $\mathbb{Z}^3$ with sixteen possible step vectors, but they are not all unit vectors. Our construction is a small variation on theirs. Also see [8], which gave a lower bound on the length of the longest good sequence in dimension k (now made obsolete for $k \geq 16$).

2 The construction

We will show that the sequence

$$\mathbf{b} = 1, 5, 6, 9, 6, 10, 11, 13, 6, 10, 11, 14, 11, 15, 12, 1, 6, 10, 11, 14, 11, 15, 12, 2, 11, 15, 12, 3, 12, 0, 1, 5, \dots$$

is good. This sequence can be defined as the fixed point of the following morphism:

Letter	Image	Letter	Image
0	1 4	8	11 12
1	1 5	9	11 13
2	1 6	10	11 14
3	1 7	11	11 15
4	6 8	12	12 0
5	6 9	13	12 1
6	6 10	14	12 2
7	6 11	15	12 3

Table 1: The morphism generating the good sequence $\mathbf{b}$.

For $n \in \mathbb{N}$, let $s_2(n)$ denote the number of 1s in the binary expansion of n , and let $t_n = s_2(n) \pmod{4}$. This is sequence [A179868](#) in the OEIS [11]. Set

$$\Gamma = \{0, 1, 2, 3\}^2, \quad a_n = (t_n, t_{n+1}), \quad \mathbf{a} = a_0 a_1 a_2 \cdots \quad (1)$$

Each ordered pair is *one letter* of the sixteen-letter alphabet Γ . For example,

$$\mathbf{a} = (0, 1)(1, 1)(1, 2)(2, 1)(1, 2)(2, 2)(2, 3)(3, 1) \cdots$$

Now form the desired good sequence

$$\mathbf{b} = 1, 5, 6, 9, 6, 10, 11, 13, 6, 10, 11, \dots$$

by applying the coding

$$\rho(r, s) = 4r + s \quad (2)$$

to each element of $\mathbf{a}$.

Appending a binary digit gives

$$s_2(2n) = s_2(n), \quad s_2(2n+1) = s_2(n) + 1.$$

Hence

$$t_{2n} = t_n, \quad t_{2n+1} = t_n + 1 \pmod{4}. \quad (3)$$

For $r \in \{0, 1, 2, 3\}$, write r^+ for the representative of $r + 1 \pmod{4}$ in that set.

Proposition 2. *The word $\mathbf{a}$ is the fixed point starting with $(0, 1)$ of the 2-uniform morphism*

$$H(r, s) = (r, r^+)(r^+, s). \quad (4)$$

Proof. A morphism acts on words by replacing each letter with its image and concatenating the resulting images. Each image in (4) consists of two letters, so H is 2-uniform. By (3),

$$\begin{aligned} a_{2n} &= (t_{2n}, t_{2n+1}) = (t_n, t_n^+), \\ a_{2n+1} &= (t_{2n+1}, t_{2n+2}) = (t_n^+, t_{n+1}). \end{aligned}$$

Thus $H(a_n) = a_{2n}a_{2n+1}$ for every n , and concatenation gives $H(\mathbf{a}) = \mathbf{a}$.

Also $t_0 = 0$, $t_1 = 1$, and $H(0, 1) = (0, 1)(1, 1)$. Therefore $H^{j+1}(0, 1)$ begins with $H^j(0, 1)$, whose length is 2^j . These prefixes determine a unique infinite word. Since $\mathbf{a}$ is such a fixed point, it is precisely this limit. $\square$

Finally, applying the coding ρ rewrites the morphism as

$$(4r + s) \rightarrow (4r + r^+)(4r^+ + s), \quad (5)$$

which gives us the morphism in Table 1.

3 The arithmetic ingredient

For a nonzero integer v , define the 2-adic valuation $\nu_2(v)$ by writing $v = 2^e w$, with w odd. Then $\nu_2(v) = e$. For a positive rational number a/b , define

$$\nu_2(a/b) = \nu_2(a) - \nu_2(b).$$

This is independent of the chosen representation, because $a/b = c/d$ implies $ad = bc$, and the exponent of 2 in a product is the sum of the exponents in its factors. In particular,

$$\nu_2(xy) = \nu_2(x) + \nu_2(y), \quad \nu_2(x/y) = \nu_2(x) - \nu_2(y). \quad (6)$$

Let $i^2 = -1$, and put

$$u_n = i^{t_n} = i^{s_2(n)}, \quad Z_0 = 0, \quad Z_n = \sum_{j=0}^{n-1} u_j \quad (n \geq 1). \quad (7)$$

The equality of the two powers of i uses $i^4 = 1$. Each u_j is one of $1, i, -1, -i$, so each Z_n is a Gaussian integer. For a Gaussian integer $X + iY$, its squared modulus is the ordinary nonnegative integer $X^2 + Y^2$.

Equation (3) gives

$$u_{2n} = u_n, \quad u_{2n+1} = iu_n.$$

Pairing consecutive terms in the defining sum for Z_{2n} , we obtain

$$\begin{aligned} Z_{2n} &= \sum_{j=0}^{n-1} (u_{2j} + u_{2j+1}) = \sum_{j=0}^{n-1} (1+i)u_j = (1+i)Z_n, \\ Z_{2n+1} &= Z_{2n} + u_{2n} = (1+i)Z_n + u_n. \end{aligned}$$

Equivalently, for $\varepsilon \in \{0, 1\}$,

$$u_{2n+\varepsilon} = i^\varepsilon u_n, \quad Z_{2n+\varepsilon} = (1+i)Z_n + \varepsilon u_n. \quad (8)$$

The following lemma is from [4].

Lemma 3. *If $0 \leq m < n$ and $t_m = t_n$, then $Z_n - Z_m \neq 0$ and*

$$\nu_2(|Z_n - Z_m|^2) = \nu_2(n - m). \quad (9)$$

Proof. Let $e = \nu_2(n - m)$. We first explain one halving step. If $n - m$ is even, the indices have the same parity, so write

$$m = 2a + \varepsilon, \quad n = 2b + \varepsilon, \quad \varepsilon \in \{0, 1\}.$$

The hypothesis $t_m = t_n$ is equivalent to $u_m = u_n$. The first identity in (8) then gives $i^\varepsilon u_a = i^\varepsilon u_b$, hence $u_a = u_b$ and $t_a = t_b$. The second identity consequently gives

$$\begin{aligned} Z_n - Z_m &= (1+i)(Z_b - Z_a) + \varepsilon(u_b - u_a) \\ &= (1+i)(Z_b - Z_a). \end{aligned}$$

After exactly e such steps, there are indices $m' < n'$ with odd difference and

$$Z_n - Z_m = (1+i)^e (Z_{n'} - Z_{m'}). \quad (10)$$

If $e = 0$, this means $m' = m$ and $n' = n$.

Write $Z_{n'} - Z_{m'} = X + iY$. This difference is a sum of the odd number $n' - m'$ of Gaussian units u_j . The sum of the real and imaginary coordinates of each unit is either 1 or -1 , both odd. It follows that $X + Y$ is odd. Since an integer and its square have the same parity,

$$X^2 + Y^2 \equiv X + Y \equiv 1 \pmod{2}.$$

In particular, this squared modulus is positive and odd. Taking squared moduli in (10), and using $|1+i|^2 = 2$, gives

$$|Z_n - Z_m|^2 = 2^e (X^2 + Y^2).$$

The right side is nonzero and has exponent of 2 equal to e . This proves both assertions. $\square$

For a letter $(r, s) \in \Gamma$ and a finite block x of $\mathbf{a}$, now write

$$C_x(r, s) = |x|_{(r, s)}.$$

Let e_0, e_1, e_2, e_3 be the standard basis of $\mathbb{N}^4$. To a pair (r, s) assign the vector $e_r - e_s$.

For indices $\alpha < \beta$ and the block $x = a_\alpha \cdots a_{\beta-1}$, summing these vectors gives

$$\begin{aligned} \sum_{r, s=0}^3 C_x(r, s)(e_r - e_s) &= \sum_{j=\alpha}^{\beta-1} (e_{t_j} - e_{t_{j+1}}) \\ &= e_{t_\alpha} - e_{t_\beta}. \end{aligned} \tag{11}$$

Lemma 4. *Suppose consecutive blocks $x = a_\alpha \cdots a_{\beta-1}$ and $y = a_\beta \cdots a_{\gamma-1}$, with $\alpha < \beta < \gamma$, have the same normalized Parikh vectors. Then $t_\alpha = t_\beta = t_\gamma$.*

Proof. Set $p = \beta - \alpha$ and $q = \gamma - \beta$. The hypothesis says

$$qC_x(r, s) = pC_y(r, s) \quad (0 \leq r, s \leq 3). \tag{12}$$

Multiply each equality by $e_r - e_s$, sum, and apply (11) to the two blocks. The result is

$$q(e_{t_\alpha} - e_{t_\beta}) = p(e_{t_\beta} - e_{t_\gamma}),$$

which rearranges to

$$qe_{t_\alpha} + pe_{t_\gamma} = (p + q)e_{t_\beta}. \tag{13}$$

Look at the coordinate labeled t_β . Its value on the left is

$$q \mathbf{1}_{\{t_\alpha=t_\beta\}} + p \mathbf{1}_{\{t_\gamma=t_\beta\}},$$

and its value on the right is $p + q$. Because p and q are both strictly positive, equality requires both indicators to be 1. $\square$

4 Proof that the sequence is good

Theorem 5. *The infinite word $\mathbf{a}$ defined by (1) is good. Equivalently, the standard-basis walk in $\mathbb{N}^{16}$ obtained from $\rho(a_n) = 4t_n + t_{n+1}$ has no three collinear vertices.*

Proof. Assume, to get a contradiction, that consecutive nonempty blocks x, y have identical normalized Parikh vectors. Write their positions as $[\alpha, \beta)$ and $[\beta, \gamma)$, and set $p = \beta - \alpha > 0$, $q = \gamma - \beta > 0$. Equation (12) holds, and Lemma 4 gives

$$t_\alpha = t_\beta = t_\gamma. \tag{14}$$

Assign (r, s) the complex number i^r . Since the first coordinate of a_j is t_j , grouping gives

$$\begin{aligned}\sum_{r,s=0}^3 C_x(r, s)i^r &= \sum_{j=\alpha}^{\beta-1} i^{t_j} = Z_\beta - Z_\alpha, \\ \sum_{r,s=0}^3 C_y(r, s)i^r &= \sum_{j=\beta}^{\gamma-1} i^{t_j} = Z_\gamma - Z_\beta.\end{aligned}$$

Consequently, multiplying (12) by i^r and summing yields

$$q(Z_\beta - Z_\alpha) = p(Z_\gamma - Z_\beta).$$

Put $X = Z_\beta - Z_\alpha$ and $Y = Z_\gamma - Z_\beta$. There is a common complex number $V = X/p = Y/q$. Adding the identities $X = pV$, $Y = qV$ shows that

$$\frac{X}{p} = \frac{Y}{q} = \frac{X+Y}{p+q} = V. \quad (15)$$

By (14), Lemma 3 applies to each of the three pairs of indices (α, β) , (β, γ) , and (α, γ) . Therefore

$$\begin{aligned}\nu_2(|X|^2) &= \nu_2(p), \\ \nu_2(|Y|^2) &= \nu_2(q), \\ \nu_2(|X+Y|^2) &= \nu_2(p+q).\end{aligned} \quad (16)$$

The same lemma also proves that all three displacements are nonzero. In particular $|V|^2$ is a positive rational number, so its valuation is defined.

Using the first expression for V in (15), followed by (6) and (16), gives

$$\begin{aligned}\nu_2(|V|^2) &= \nu_2\left(\frac{|X|^2}{p^2}\right) = \nu_2(|X|^2) - 2\nu_2(p) \\ &= \nu_2(p) - 2\nu_2(p) = -\nu_2(p).\end{aligned}$$

Using the other two expressions for V in exactly the same way gives

$$\nu_2(|V|^2) = -\nu_2(q) = -\nu_2(p+q).$$

Thus

$$\nu_2(p) = \nu_2(q) = \nu_2(p+q). \quad (17)$$

Let the common value of the first two terms be d . Then $p = 2^d p_0$ and $q = 2^d q_0$ for positive odd integers p_0, q_0 . Their sum is even, so

$$p+q = 2^d(p_0+q_0) \implies \nu_2(p+q) \geq d+1,$$

contradicting (17). No such blocks x, y can exist.

Relabeling letters by the bijection ρ preserves all frequency equalities. Proposition 1 now gives the asserted geometric conclusion. $\square$

5 The dimension bound and the proposed smaller morphisms

Define $k_{\min}$ to be the least size of an alphabet admitting a good infinite word. Theorem 5 proves that this minimum exists and is at most 16.

As mentioned above, we know that $k_{\min} \geq 4$. Therefore it is still of interest to determine its exact value.

I found a number of candidate cyclic morphisms, but have been unable to prove that they work. I list them here in the hope that they may aid other researchers to improve the bound $k_{\min} \leq 16$.

Alphabet size k	Proposed image of 0	Image length
5	01213101314310	14
6	01210140	8
7	012520	6
8	012560	6

For these proposals, the image of letter r is obtained by adding r modulo k to every letter in the displayed image of 0.

6 Declaration of AI usage

I originally proposed this problem about walks on unit vectors in June 2018, and mentioned it to many people. The recent construction of Cambie and Kalviainen, and the release of GPT-6 Astra, led me to try this new LLM on the unit vector version of the problem. Much of this paper was written by ChatGPT 6 Astra.

References

- [1] Sergey Avgustinovich and Svetlana Puzynina. Weak abelian periodicity of infinite words. *Theory of Computing Systems* **59** (2016), 161–179.
- [2] T. C. Brown. Is there a sequence on four symbols in which no two adjacent segments are permutations of one another? *Amer. Math. Monthly* **78** (1971), 886–888.
- [3] T. C. Brown. Advanced problem 5811. *Amer. Math. Monthly* **78** (1971), 798.
- [4] Stijn Cambie and Erik Kalviainen. An infinite small-step $\mathbb{Z}^3$ -walk with no collinear triple. ArXiv preprint arXiv:2609.01766 [math.CO], September 1 2026. Available at <https://arxiv.org/abs/2609.01766v1>.

- [5] Gabriele Fici and Svetlana Puzynina. Abelian combinatorics on words: A survey. *Computer Science Review* **47** (2023), article 100532.
- [6] Joseph L. Gerver and L. Thomas Ramsey. On certain sequences of lattice points. *Pacific Journal of Mathematics* **83** (1979), 357–363.
- [7] Samuel Korsky. North-east lattice paths with few collinear vertices. Arxiv preprint arXiv:2607.02832 [math.CO], July 2 2026. Available at <https://arxiv.org/abs/2607.02832>.
- [8] Samuel Korsky. Long lattice paths with no three collinear vertices. Arxiv preprint arXiv:2608.07906 [math.CO], August 8 2026. Available at <https://arxiv.org/abs/2608.07906>.
- [9] Thomas F. Lidbetter. Improved bound for the Gerver–Ramsey collinearity problem. *Discrete Mathematics* **347** (2024), article 113718.
- [10] P. L. Montgomery. Solution to Advanced Problem 5811: collinear points on a monotonic polygon. *American Mathematical Monthly* **79** (1972), 1143–1144.
- [11] N. J. A. Sloane et al. The On-Line Encyclopedia of Integer Sequences. Electronic resource, 2026. Available at <https://oeis.org>.